

Multi-Router Network Design & Connectivity Verification

A six-router redundant topology built and tested in Cisco Packet Tracer, with end-to-end reachability confirmed across every segment.

Project Overview

This lab sets up a small enterprise-style WAN made up of six Cisco 2621XM routers connecting three end hosts across separate subnets. The goal was to design an addressing scheme, wire up the topology, configure routing, and then prove that every host can reach every other host and every router interface in the network. The layout intentionally includes a redundant path between the core (R2) and the far side of the network (R5), routed through two parallel routers, R3 and R4, so traffic has more than one way to get across.

Topology

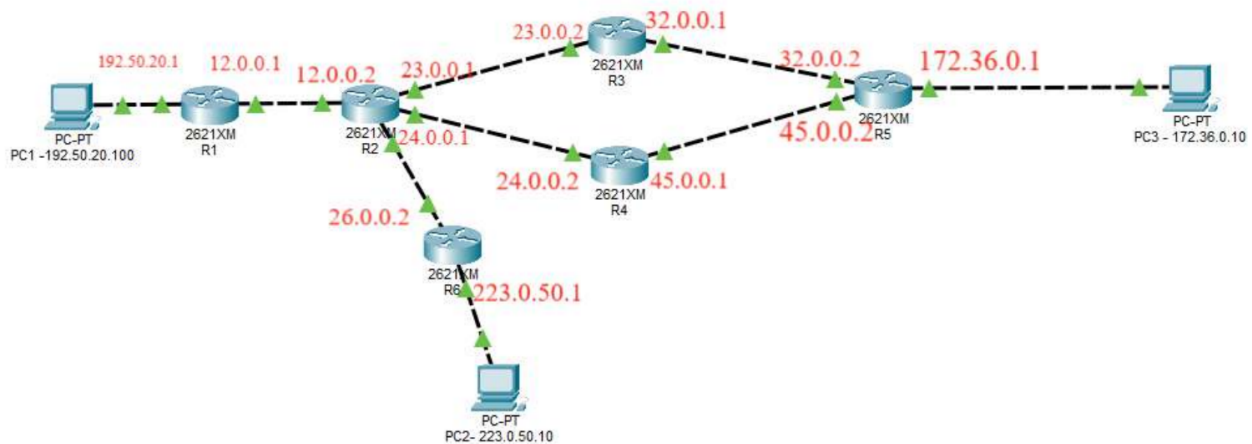


Figure 1 — Logical topology with interface IP addresses. R1 and R6 sit at the network edge with directly attached hosts; R2 is the central distribution router; R3 and R4 form a redundant pair of paths between R2 and R5.

PC1, PC2, and PC3 each sit on their own edge subnet, off R1, R6, and R5 respectively. Every router-to-router link uses its own /24 point-to-point subnet, which keeps the addressing simple to follow and easy to troubleshoot.

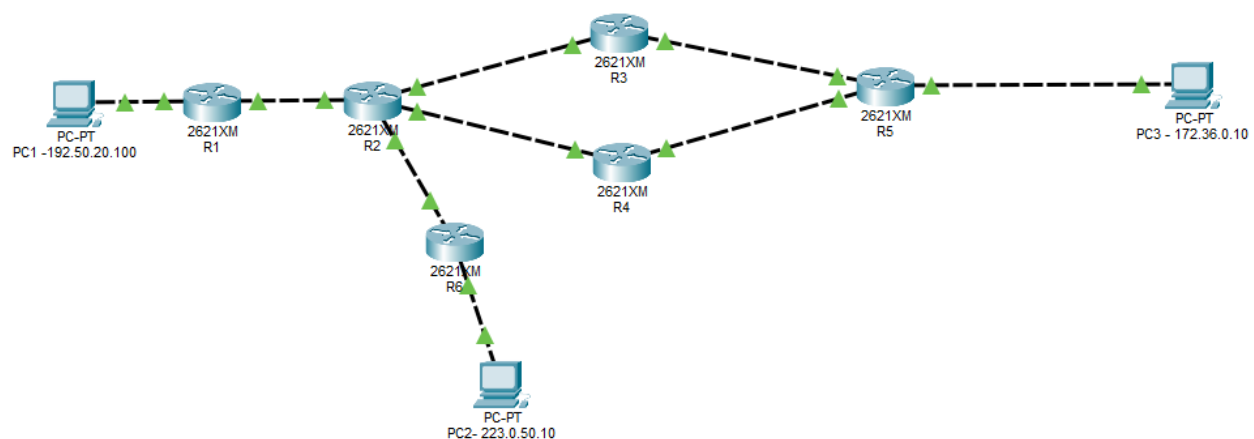


Figure 2 — Same topology, unlabeled, showing just the physical layout: six routers, three end hosts, and the redundant R2–R3–R5 / R2–R4–R5 path.

IP Addressing Plan

Device	Interface / Link	IP Address
PC1	LAN (gateway: R1)	192.50.20.100
R1	LAN (to PC1)	192.50.20.1
R1	to R2	12.0.0.1
R2	to R1	12.0.0.2
R2	to R3	23.0.0.1
R2	to R4	24.0.0.1
R2	to R6	26.0.0.1
R3	to R2	23.0.0.2
R3	to R5	32.0.0.1
R4	to R2	24.0.0.2
R4	to R5	45.0.0.1
R5	to R3	32.0.0.2
R5	to R4	45.0.0.2
R5	LAN (to PC3)	172.36.0.1
R6	to R2	26.0.0.2
R6	LAN (to PC2)	223.0.50.1
PC2	LAN (gateway: R6)	223.0.50.10
PC3	LAN (gateway: R5)	172.36.0.10

Connectivity Verification

Once addressing and routing were in place, I ran ping tests from all three end hosts to a mix of nearby and far-side destinations — router interfaces, the redundant R3/R4 links, and the opposite LANs. Every test came back 0% loss, which confirms the routing tables are converged correctly on all six routers and both paths through R3 and R4 are usable.

From	To	Destination	Result
PC1 (192.50.20.100)	R1 LAN gateway	192.50.20.1	0% loss
PC1	R2 (near side)	12.0.0.1	0% loss
PC1	R2 (far side)	12.0.0.2	0% loss
PC1	R6 (via R2)	26.0.0.2	0% loss
PC1	PC2 (edge-to-edge)	223.0.50.10	0% loss
PC1	R6 LAN gateway	223.0.50.1	0% loss
PC1	R5, via R4 path	45.0.0.2	0% loss
PC1	R5, via R3 path	32.0.0.2	0% loss
PC1	PC3 (edge-to-edge)	172.36.0.10	0% loss
PC2 (223.0.50.10)	PC1	192.50.20.100	0% loss
PC2	PC3	172.36.0.10	0% loss
PC2	R3 (far side)	32.0.0.1	0% loss
PC3 (172.36.0.10)	PC2	223.0.50.10	0% loss
PC3	PC1	192.50.20.100	0% loss
PC3	R2 (across the core)	24.0.0.1	0% loss

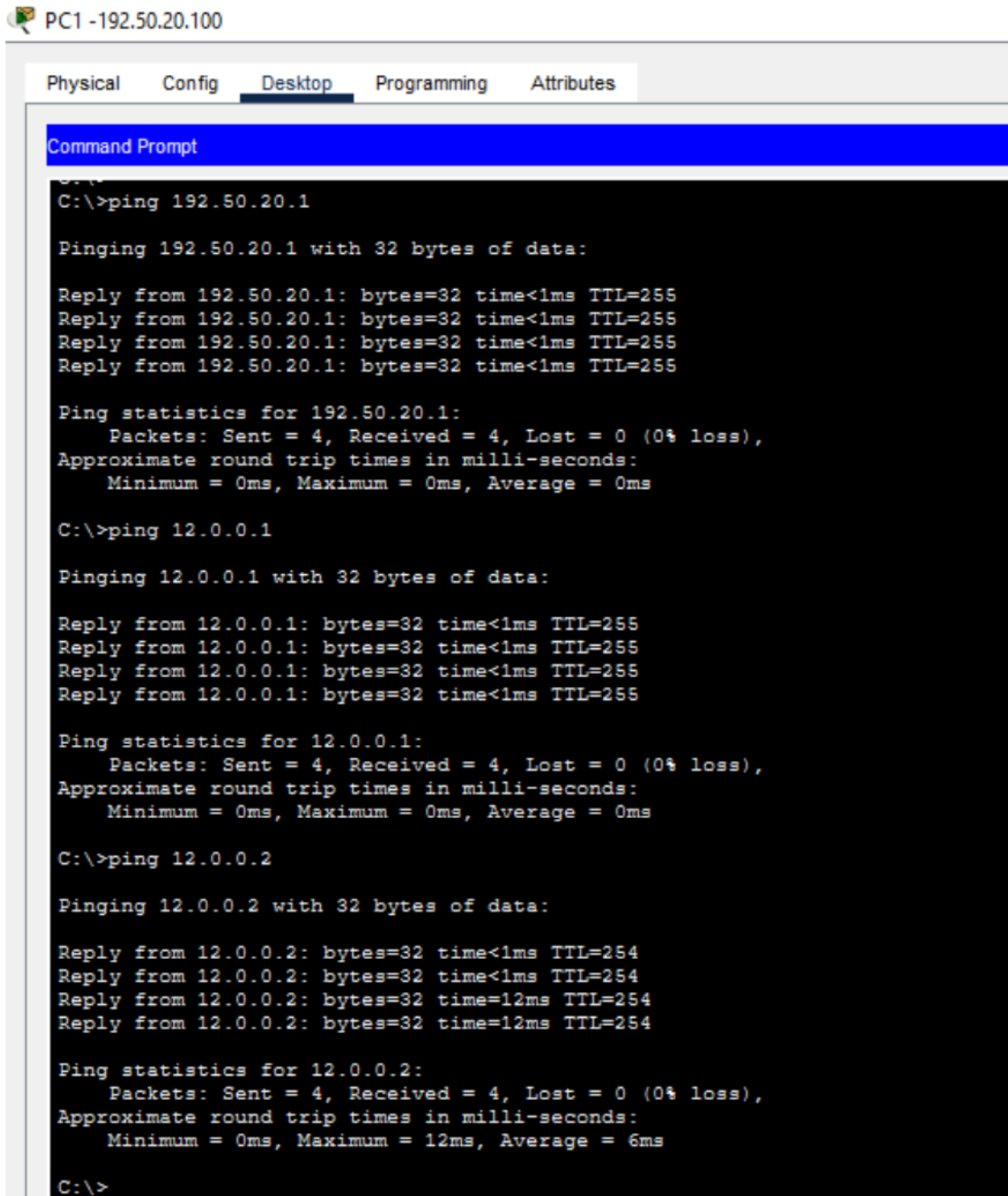
A quick note on TTL values while reading raw ping output: the decreasing TTL on replies (e.g. 255 → 254 → 253 for PC1 reaching R1, then R2, then R6) simply reflects hop count, and it lines up exactly with the path each packet should be taking through the topology — a good sanity check that traffic is following the expected route rather than something unintended.

Takeaways

The main things this lab reinforced for me: keeping a clean, predictable addressing scheme makes a topology far easier to reason about once it grows past three or four routers, and building in redundant paths (R3/R4 here) only pays off if the routing protocol is actually verified end-to-end afterward — a topology diagram alone doesn't prove reachability. All six routers and three hosts were confirmed reachable from every other point in the network with zero packet loss.

Appendix — Raw Ping Output

The screenshots below are the actual Packet Tracer command-prompt sessions the results table above was drawn from, kept here for anyone who wants to see the raw output.



PC1 - 192.50.20.100

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ping 192.50.20.1

Pinging 192.50.20.1 with 32 bytes of data:

Reply from 192.50.20.1: bytes=32 time<1ms TTL=255
Reply from 192.50.20.1: bytes=32 time<1ms TTL=255
Reply from 192.50.20.1: bytes=32 time<1ms TTL=255
Reply from 192.50.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.50.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 12.0.0.1

Pinging 12.0.0.1 with 32 bytes of data:

Reply from 12.0.0.1: bytes=32 time<1ms TTL=255
Reply from 12.0.0.1: bytes=32 time<1ms TTL=255
Reply from 12.0.0.1: bytes=32 time<1ms TTL=255
Reply from 12.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 12.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 12.0.0.2

Pinging 12.0.0.2 with 32 bytes of data:

Reply from 12.0.0.2: bytes=32 time<1ms TTL=254
Reply from 12.0.0.2: bytes=32 time<1ms TTL=254
Reply from 12.0.0.2: bytes=32 time=12ms TTL=254
Reply from 12.0.0.2: bytes=32 time=12ms TTL=254

Ping statistics for 12.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 6ms

C:\>
```

Figure — PC1 → 192.50.20.1, 12.0.0.1, 12.0.0.2

Physical Config **Desktop** Programming Attributes

Command Prompt

```
C:\>ping 26.0.0.2
```

```
Pinging 26.0.0.2 with 32 bytes of data:
```

```
Reply from 26.0.0.2: bytes=32 time<1ms TTL=253
Reply from 26.0.0.2: bytes=32 time<1ms TTL=253
Reply from 26.0.0.2: bytes=32 time=13ms TTL=253
Reply from 26.0.0.2: bytes=32 time<1ms TTL=253
```

```
Ping statistics for 26.0.0.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

```
C:\>ping 223.0.50.10
```

```
Pinging 223.0.50.10 with 32 bytes of data:
```

```
Reply from 223.0.50.10: bytes=32 time<1ms TTL=125
Reply from 223.0.50.10: bytes=32 time=1ms TTL=125
Reply from 223.0.50.10: bytes=32 time<1ms TTL=125
Reply from 223.0.50.10: bytes=32 time<1ms TTL=125
```

```
Ping statistics for 223.0.50.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

```
C:\>ping 223.0.50.1
```

```
Pinging 223.0.50.1 with 32 bytes of data:
```

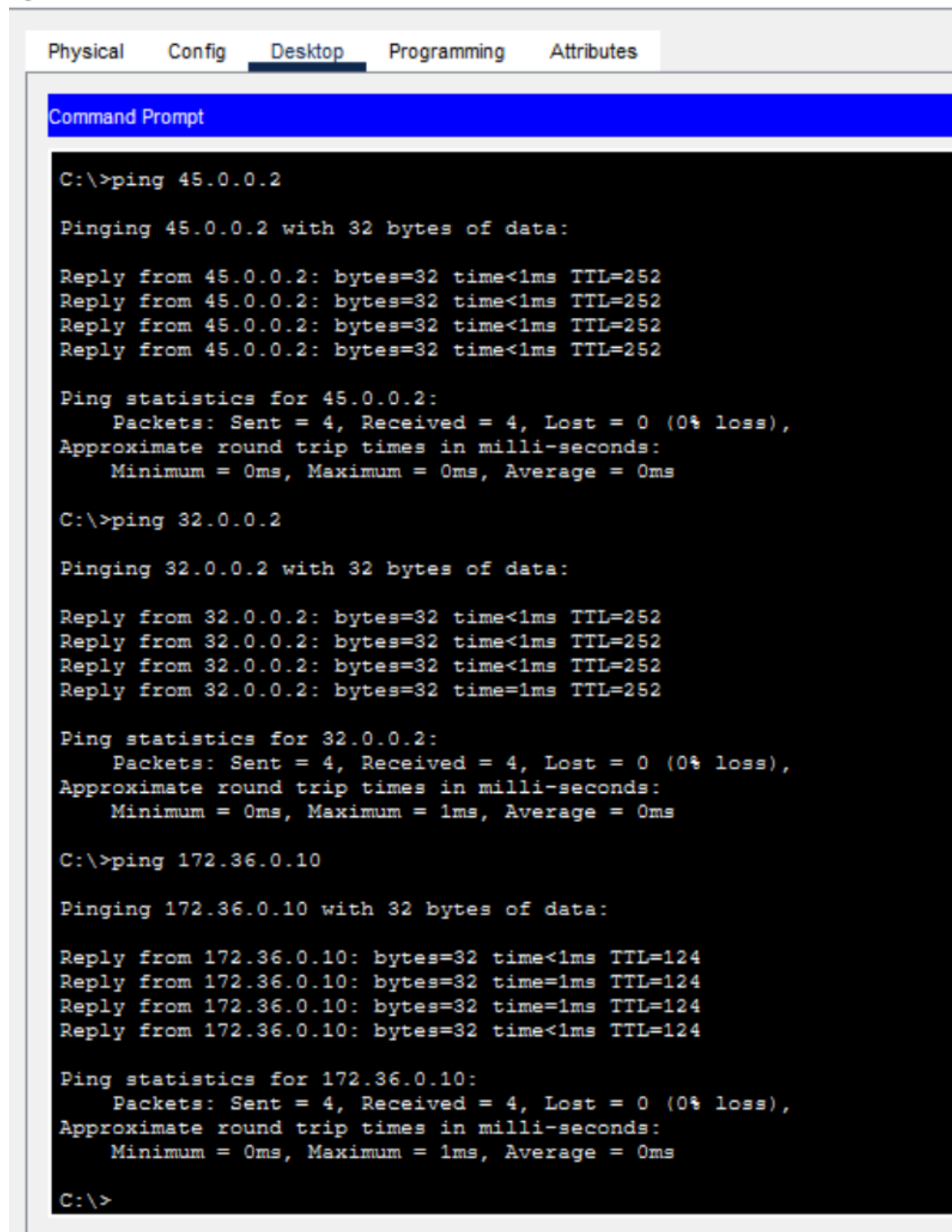
```
Reply from 223.0.50.1: bytes=32 time<1ms TTL=253
Reply from 223.0.50.1: bytes=32 time<1ms TTL=253
Reply from 223.0.50.1: bytes=32 time<1ms TTL=253
Reply from 223.0.50.1: bytes=32 time<1ms TTL=253
```

```
Ping statistics for 223.0.50.1:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>
```

Figure — PC1 → 26.0.0.2, 223.0.50.10, 223.0.50.1



The screenshot shows a desktop environment with a taskbar at the top containing icons for Physical, Config, Desktop (selected), Programming, and Attributes. Below the taskbar is a Command Prompt window with a blue title bar labeled 'Command Prompt'. The window displays the results of three ping commands executed from the C:\ prompt.

```
C:\>ping 45.0.0.2

Pinging 45.0.0.2 with 32 bytes of data:

Reply from 45.0.0.2: bytes=32 time<1ms TTL=252
Reply from 45.0.0.2: bytes=32 time<1ms TTL=252
Reply from 45.0.0.2: bytes=32 time<1ms TTL=252
Reply from 45.0.0.2: bytes=32 time<1ms TTL=252

Ping statistics for 45.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 32.0.0.2

Pinging 32.0.0.2 with 32 bytes of data:

Reply from 32.0.0.2: bytes=32 time<1ms TTL=252
Reply from 32.0.0.2: bytes=32 time<1ms TTL=252
Reply from 32.0.0.2: bytes=32 time<1ms TTL=252
Reply from 32.0.0.2: bytes=32 time=1ms TTL=252

Ping statistics for 32.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.36.0.10

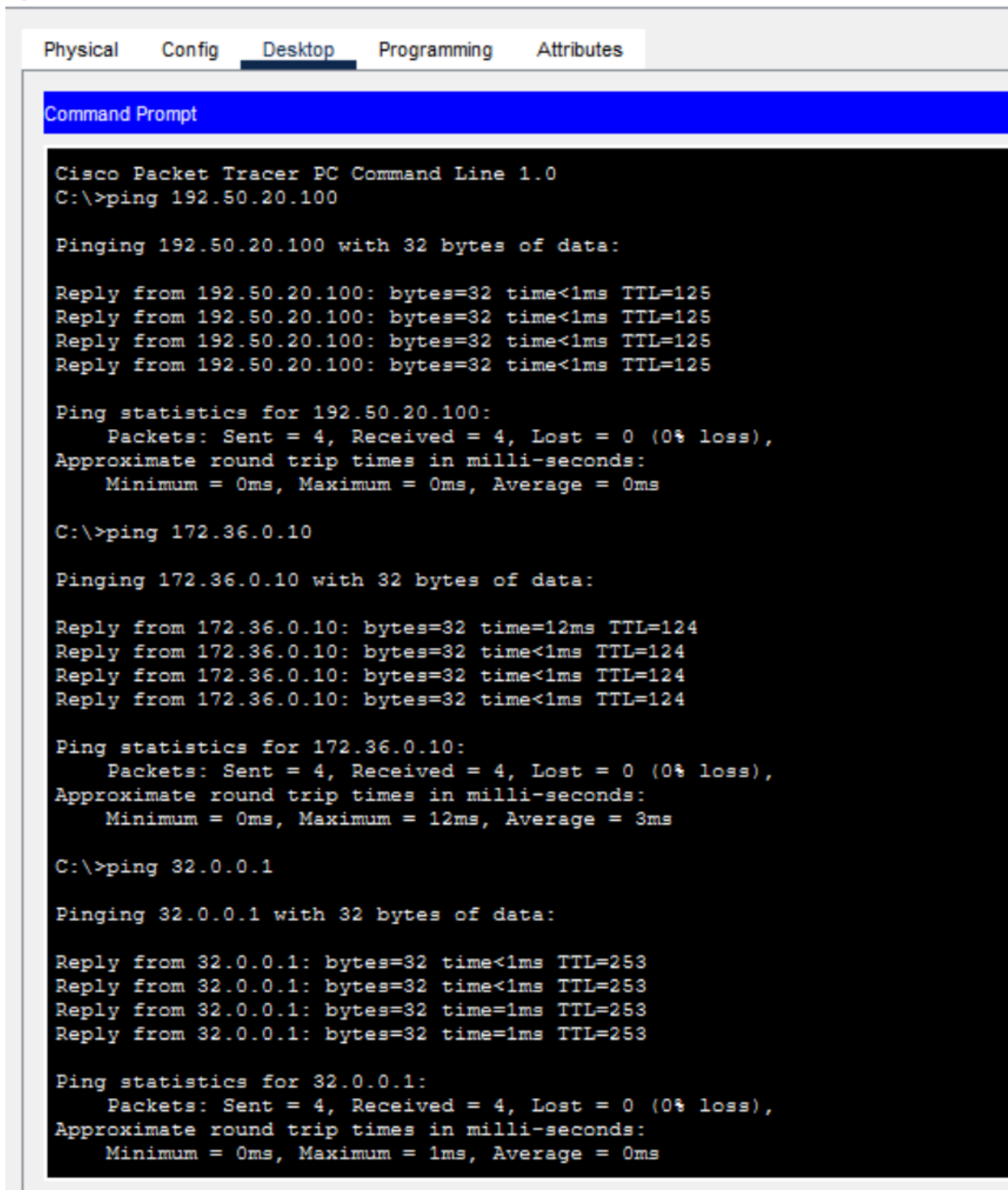
Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time<1ms TTL=124
Reply from 172.36.0.10: bytes=32 time=1ms TTL=124
Reply from 172.36.0.10: bytes=32 time=1ms TTL=124
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure — PC1 → 45.0.0.2, 32.0.0.2, 172.36.0.10



The image shows a Cisco Packet Tracer PC Command Line interface for PC2 with IP 223.0.50.10. The interface has tabs for Physical, Config, Desktop, Programming, and Attributes. The Desktop tab is active, displaying a Command Prompt window. The Command Prompt shows the execution of three ping commands from the PC's perspective. Each command is followed by a status message, four individual replies with their respective byte counts, times, and TTLs, and then a summary of ping statistics including packets sent/received/lost and round trip times.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time<1ms TTL=125
Reply from 192.50.20.100: bytes=32 time<1ms TTL=125
Reply from 192.50.20.100: bytes=32 time<1ms TTL=125
Reply from 192.50.20.100: bytes=32 time<1ms TTL=125

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=12ms TTL=124
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124
Reply from 172.36.0.10: bytes=32 time<1ms TTL=124

Ping statistics for 172.36.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ping 32.0.0.1

Pinging 32.0.0.1 with 32 bytes of data:

Reply from 32.0.0.1: bytes=32 time<1ms TTL=253
Reply from 32.0.0.1: bytes=32 time<1ms TTL=253
Reply from 32.0.0.1: bytes=32 time=1ms TTL=253
Reply from 32.0.0.1: bytes=32 time=1ms TTL=253

Ping statistics for 32.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure — PC2 → 192.50.20.100, 172.36.0.10, 32.0.0.1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=12ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124
Reply from 223.0.50.10: bytes=32 time<1ms TTL=124
Reply from 223.0.50.10: bytes=32 time<1ms TTL=124

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ping 192.50.20.100

Pinging 192.50.20.100 with 32 bytes of data:

Reply from 192.50.20.100: bytes=32 time<1ms TTL=124
Reply from 192.50.20.100: bytes=32 time=1ms TTL=124
Reply from 192.50.20.100: bytes=32 time=1ms TTL=124
Reply from 192.50.20.100: bytes=32 time<1ms TTL=124

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 24.0.0.1

Pinging 24.0.0.1 with 32 bytes of data:

Reply from 24.0.0.1: bytes=32 time=13ms TTL=253
Reply from 24.0.0.1: bytes=32 time<1ms TTL=253
Reply from 24.0.0.1: bytes=32 time=2ms TTL=253
Reply from 24.0.0.1: bytes=32 time<1ms TTL=253

Ping statistics for 24.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 13ms, Average = 3ms
```

Figure — PC3 → 223.0.50.10, 192.50.20.100, 24.0.0.1